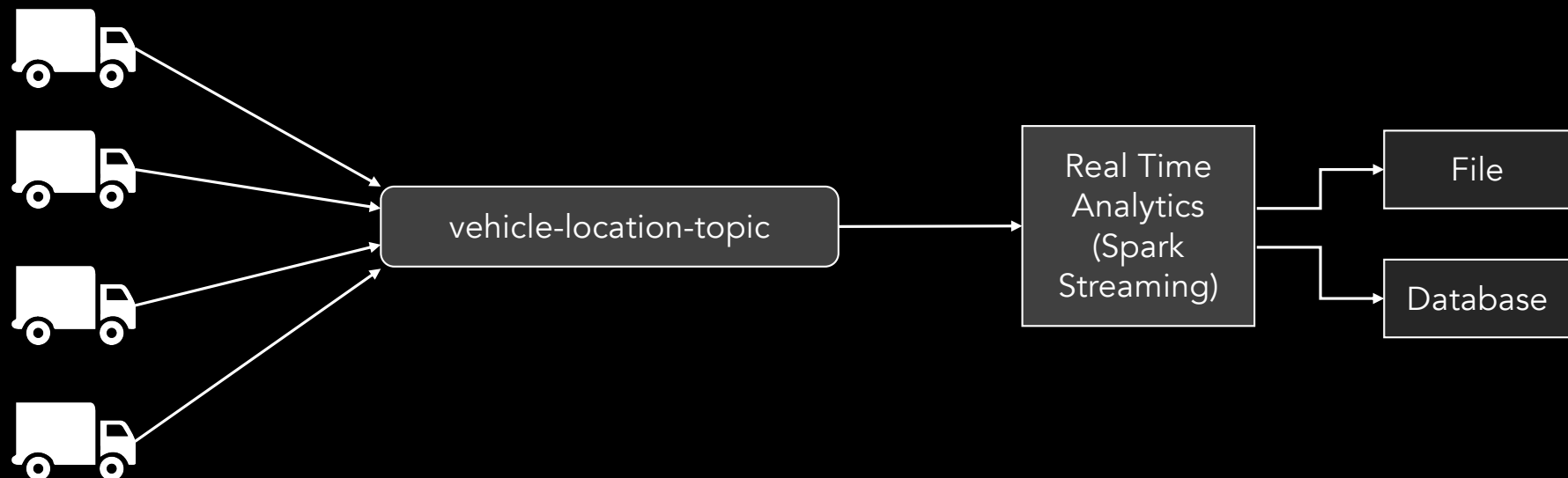


Project 9 – Vehicle Route Analysis

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Vehicle Route Analysis

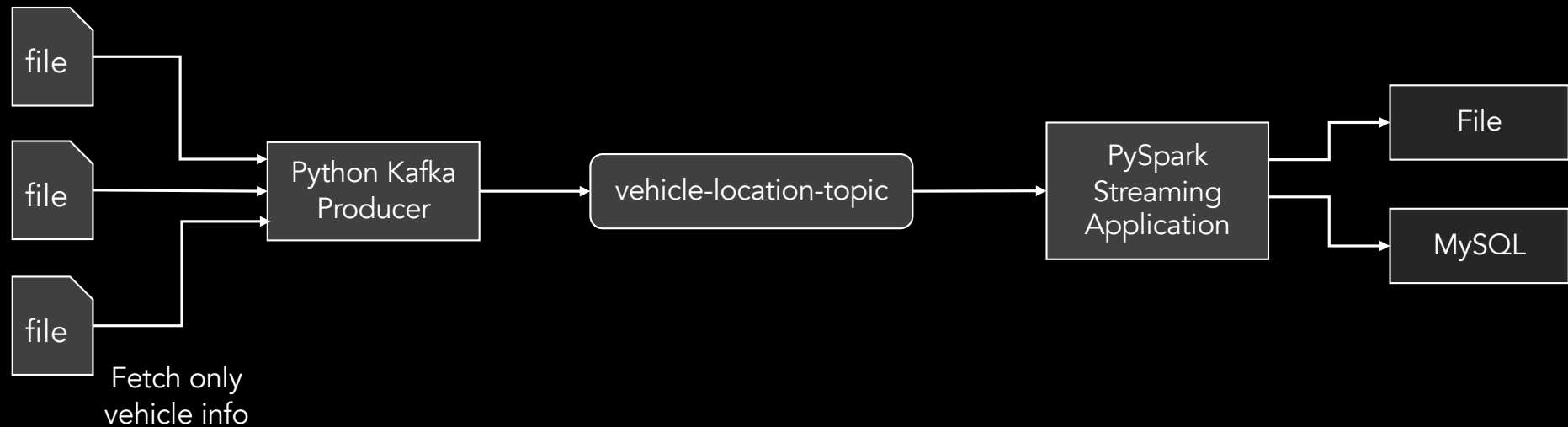
- JSON Dataset Size (uncompressed) – 17.34 GB
- Total no. of files – 50 (1 for each vehicle).
- Exercise for Python and PySpark Streaming.



Route Coverage

- Find out the total distance covered by each vehicle every 1 hour for near real-time dashboard.
- Vehicle no. 28 is transporting critical material for power station at a rural area on 2026-01-06. Material is picked at 3am and scheduled to be dropped at 3pm on the same day covering 1000KM. Need to continuously monitor the avg distance covered by the vehicle, every 15 seconds, to check whether it is on time or not and flag whenever the avg speed is below the needed speed. Consider a buffer of 1-2 hrs end-to-end, data older than 2 hrs can be ignored.
 - Note : Use latitude and longitude together to find the distance travelled from point A (starting at 3am) to point B (ending at 3pm). Latitude and longitude are in Degrees.
 - Note : Use Haversine formula to find distance between two pairs of LatLong. It is used to find distance between two points in a sphere. Python implementation link - <https://www.geeksforgeeks.org/dsa/haversine-formula-to-find-distance-between-two-points-on-a-sphere/>.
 - Check the output of Haversine Python implementation here - <https://www.omnicalculator.com/other/latitude-longitude-distance>
- Use Python multi-threading to read multiple files together and send data points to Kafka.
- 1 Degree Latitude = 100 km. 1 Degree Longitude = 100 km x cos(latitude in degree)

Route Coverage - Implementation

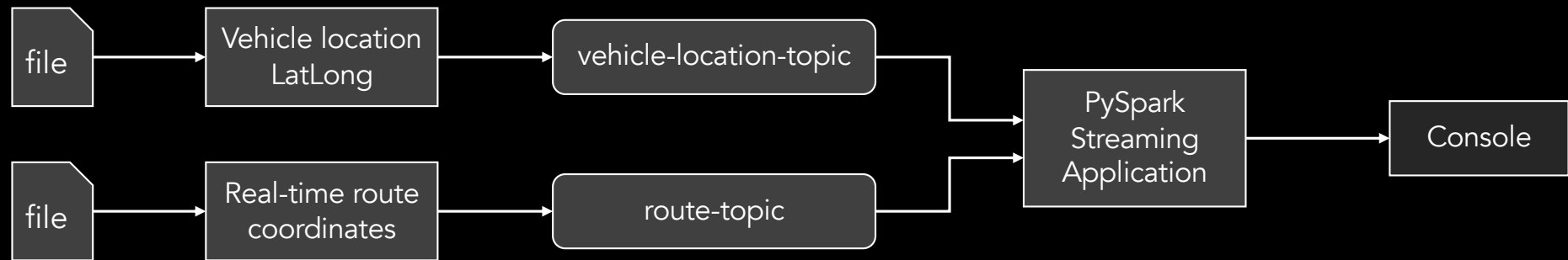


- vehicle-location-topic should have 10 – 20 partitions.
- Use python multi-threading to read multiple files using a single program and send to Kafka topic.
- Multiple programs can be used along with multi-thread to achieve high throughput.
- Output data should be stored in File and MySQL database both.

Route Deviation

- Real-time route co-ordinates (LatLong) are being sent to vehicle pilot/driver. Each vehicle should follow the route sent to vehicle pilots.
- Vehicle topic – Vehicle Id, Vehicle LatLong, Vehicle Timestamp.
- Route topic – Vehicle Id, Route LatLong, Route Timestamp.
- Find deviation of vehicle traverse from route data. If deviation is more than 30km, then flag the vehicle information in the console.
- Deviation calculation
 - $\text{LatLong deviation} = \text{Vehicle LatLong} - \text{Route LatLong}$.
 - Distance deviation = Use Haversine formula to find the distance in km.
- 1 Degree Latitude = 100 km. 1 Degree Longitude = $100 \text{ km} \times \cos(\text{latitude in degree})$

Route Deviation - Implementation



- Send vehicle location to Kafka topic and route data to route-topic.
- Use Python multi-threading to read multiple files using same code.
- Read from one input file and send data to two different topics.